



# Amphiphilic cone-shaped cationic calix[4]arene composite anion exchange membranes with continuous ionic channels

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<https://doi.org/10.1016/j.memsci.2021.119815> 

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## Highlights

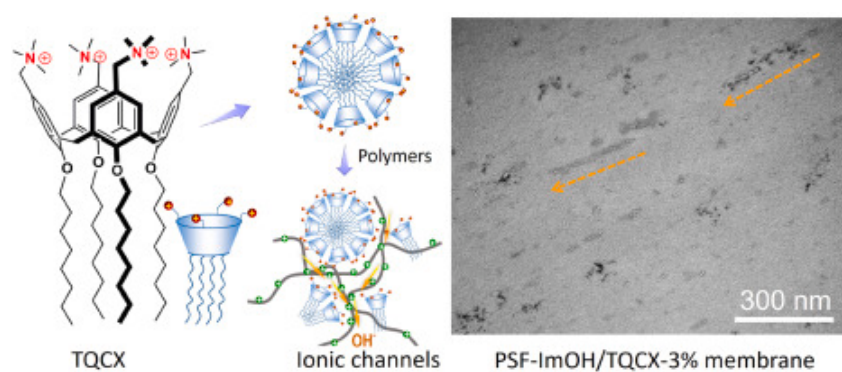
- An amphiphilic cone-shaped TQCX is proposed to enhance micro-phase separation.
- TQCX promotes ionic and hydrophobic aggregations, and also introduces free volume.
- Continuous ionic channels are efficiently constructed with the length up to 300nm.
- The highest ion conductivity is achieved at 3wt% TQCX, yielding a 70% increase.

## Abstract

A novel amphiphilic cone-shaped inducer, tetra-quaternized tetraoctyloxycalix[4]arene (TQCX), is proposed to create continuous ionic channels in an anion exchange membrane (AEM). Via molecular dynamics simulation, it was found that four upper-rim cationic groups induce ionic aggregation via electrostatic interactions, while four lower-rim hydrophobic octyl side chains promote the clustering of hydrophobic domains via van der Waals interactions with the polymer backbone. The core cavity in TQCX introduces additional free volume to connect hydrophilic

domains. At the optimal TQCX content of 3wt%, the ionic clusters with a diameter of  $\sim 12$  nm are induced, and even linearly arranged to construct continuous ionic channels with a length up to  $\sim 300$  nm. Besides, TQCX also aggregates to form the ion-rich nanoparticles with the size of  $\sim 500$  nm, which also assist in the construction of interconnected hydroxide-conducting pathways. Consequently, hydroxide conductivity of the composite membrane yields a 70% improvement at 3wt% TQCX over that of the pristine membrane, achieving the top level among the reported main-chain-type AEMs based on the polysulfone backbone.

## Graphical abstract



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## Introduction

Anion exchange membrane (AEM) fuel cells have been drawing significant attention due to the flexibility of fuels and non-platinum electrocatalysts [1,2]. Although AEM as the core component has been intensively studied, its ionic conductivity is difficult to reach a comparable level to proton counterparts (e.g., Nafion) due to tortuous ionic channels [3]. Via well-designed architectures of polymers, such as highly densely cationic groups [4], hydrophobic or hydrophilic side chains [5,6] and block copolymers [7], micro-phase separations in AEMs could be tailored, constructing well-ordered ionic channels. Polymers with intrinsic microporosity can also facilitate the efficient transport of anions by subnanometer free-volume voids [8]. Unfortunately, strong interchain interactions in multi-cationic polymers and high rigidity in microporous polymers might lead to low solubility and insufficient membrane-forming capability [9,10]. Bulky rigid cyclodextrins (CDs) were also attached to polymers to promote the formation of ionic channels via the high free volume and external hydrophilic surface [11]. However, the deprotonation of CD's hydroxyl groups in alkaline solutions might make the negative-charged CDs ionic crosslink with cationic groups in AEMs [12], limiting the mobility and sacrificing ion transport ability of these binding cationic groups.

Inorganic nanofillers have been extensively incorporated into polymer matrices to create ionic channels at polymer-filler interfaces via various non-covalent interactions [13], but a trade-off between membrane integrity and ion transport usually occurs due to their intrinsic blocking nature

and severe agglomeration. And, among various additives, ionic additives may have a greater influence on regulating ionic channels, as ion-ion interactions are stronger than other non-covalent interactions [14]. Inorganic ionic additive, i.e.,  $\text{Zn}^{2+}$  ion, has been reported as a soft template to enlarge ionic clusters in AEMs via electrostatic interactions without the incompatibility issue, as the templates were removed from the membranes [15]. Organic ionic surfactants not only have a good compatibility with polymer matrices, but also self-assemble to form micelles with desirable shapes and sizes for regulating micro-phase separation in membranes [16]. However, these water-soluble ionic additives usually suffer from the leakage issues, resulting in defects in membranes and subsequently deteriorating fuel crossover [17]. To address this issue, physical crosslinking has been adopted in hydrophobic domains, as it enhances the polymer-additive interactions and also has limited impact on hydrophilic ionic channels compared with chemical crosslinking [18]. For instance, via the hydrophobic interactions between the alkyl dodecyl side chains in the ionic additive and alkyl-substituted quaternized PPO polymers, physically-crosslinked membranes were prepared, yielding a good balance between high conductivity and low swelling ratio [19].

Inspired by the merits of CDs and physical crosslinking, a novel amphiphilic cone-shaped macrocyclic tetra-quaternized tetraoctyloxycalix[4]arene (TQCX) is proposed herein to create ionic channels in AEMs. As the third generation of supramolecular host molecules after crown ethers and CDs [20], the calix[n]arenes possess tunable 3D-shaped hydrophobic cavities similar to CDs, and also has the physical-crosslinking potential due to the readily modifications of the rims [21]. Moreover, the elaborately designed calix[n]arenes have a strong self-assembly ability. The amphiphilic cone-shaped cationic calix[4]arenes have been reported to self-assemble into the globular micelles with a diameter of  $\sim 6\text{nm}$ , which induce a hierarchical assembly of DNA nanoparticles for gene delivery [22]. As compared with the CDs derivatives applied in AEMs, TQCX in this work is synthesized by separately and selectively modification of each rim rather than non-selective functionalization in CDs, and possesses stable configuration in strong alkaline conditions. To the best of our knowledge, it is the first time that the calixarene-based inducer is applied to regulate micro-phase separation in ion exchange membranes.

In this work, TQCX is proposed to efficiently create continuous ionic channels in the imidazolium ( $\text{Im}^+$ )-functionalized polysulfone membranes (PSF- $\text{ImOH}$ ). Four upper-rim quaternary ammonium groups interact electrostatically with  $\text{Im}^+$  groups in the polymer matrix, while four lower-rim octyl side chains have van der Waals (vdW) interactions with the polysulfone backbone, enhancing hydrophilic-hydrophobic micro-phase separation. In the meantime, the core cavity of TQCX introduces additional free volume to connect ionic clusters and broaden ionic channels. At the optimal content of TQCX (3wt%), continuous ionic channels composed by the ionic clusters ( $\sim 12\text{nm}$ ) are constructed with the length up to  $\sim 300\text{nm}$ . The ion-rich TQCX aggregation with a size of  $\sim 500\text{nm}$  could further contribute to the consecutive ionic channels. Consequently, hydroxide conductivity of the composite membrane yields a 70% improvement over that of the pristine membrane, achieving the top level of the previously reported polysulfone-based main-chain-type AEMs.

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## Section snippets

### Materials

Polysulfone (PSF, Udel 3500) was purchased from Amoco Performance Product Inc.. Chloromethyl octyl ether (CMOE) was prepared according to the literature [23]. Potassium hydroxide (KOH, 98%), stannic chloride (SnCl<sub>4</sub>, AR), methylene chloride (CH<sub>2</sub>Cl<sub>2</sub>, AR), 1-methyl-2-pyrrolidone (NMP, AR), anhydrous ethanol (AR), dimethyl sulfoxide (DMSO, AR) and chloroform (CHCl<sub>3</sub>, AR) were purchased from Sinopharm chemical reagent Co., Ltd. (Shanghai, China). Calix[4]arene (CX, >98.0%) and 1-bromooctane (98.0%) ...

### Synthesis of the TQCX inducer

The amphiphilic TQCX was successfully synthesized as confirmed by <sup>1</sup>H NMR in Fig. 2. In the spectrum of bromooctane, the signal at 3.41 ppm is attributed to the protons in -CH<sub>2</sub>Br (H<sub>5</sub>). According to the integration intensity ratio, the signal at 0.88 ppm is owing to -CH<sub>3</sub> (H<sub>1</sub>), while the signals of H<sub>2</sub>, H<sub>3</sub> and H<sub>4</sub> appear at 1.28 ppm, 1.41 ppm and 1.87 ppm, respectively. After reacting with CX, the signals for H<sub>4</sub>, H<sub>3</sub>, H<sub>2</sub> and H<sub>1</sub> are almost unchanged in the spectrum of TOOCX. In contrast, the signal ...

### Conclusions

A novel amphiphilic TQCX inducer is proposed herein to construct well-ordered ionic channels for rapid ion transport in the PSF-ImOH AEMs. Via vdW and electrostatic interactions as well as free-volume effect, TQCX molecules effectively create ionic channels in the membranes. At the optimal content of TQCX (3 wt%), the size of ionic cluster increases to ~12 nm, and continuous ionic channels with the length up to ~300 nm are created. Additionally, TQCX aggregates to forms ion-rich nanoparticles ...

### Author statement

Wanting Chen: Conceptualization, Investigation, Validation, Writing-Original Draft, Visualization, Funding acquisition; Xiaozhou Wang: Investigation, Validation; Tiantian Li: Writing-Review & Editing; Xiaoming Yan: Writing-Review & Editing; Xuemei Wu: Conceptualization, Writing-Review & Editing, Funding acquisition; Yang Zhang: Visualization; Fan Zhang: Validation; Shouhai Zhang: Writing-Review & Editing; Gaohong He: Supervision, Resources, Funding acquisition. ...

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

## Acknowledgements

The authors thank the National Natural Science Foundation of China (22008021, 21776034, Creative Research Groups 22021005), China Postdoctoral Science Foundation (2020M680039), National Key Research and Development Program of China (2019YFE0119200), Fundamental Research Funds for the Central Universities (DUT21ZD406, DUT20LAB307). The authors also thank the Supercomputing Center of Dalian University of Technology for providing computing resources. ...

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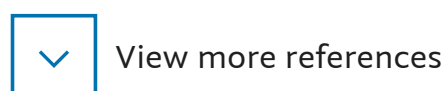
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...Cyclodextrin has hydrophobic cavities with different sizes and the rim of cyclodextrin owned many hydroxyl groups, these structural merits could be employed to build hydrophilic materials via hydrogen bonds, hydrophobic interactions, entropy effect, and inclusion interactions [100,101]. Calixarene is composed of methylene bridged phenol units and could provide abundant electrostatic interactions, hydrophobic interactions,  $\pi$ - $\pi$  interactions, hydrogen bonds, inclusion interactions, C-H... $\pi$ , cation... $\pi$  interactions to constructing rotaxanes and pseudo-rotaxanes which could assemble into functional supramolecular polymers [102,103]. Cucurbit[n]uril possesses pumpkin shape structure and electron-rich cavity with variable size, which could accommodate kinds of guest molecules via entropy effect, hydrogen bonds, inclusion interactions, ion-dipole interactions, electrostatic interactions, hydrophobic interactions,  $\pi$ - $\pi$  interactions [104,105]...

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...The membrane exhibited a high ionic conductivity ( $93.2 \text{ mS cm}^{-1}$ ), a low swelling rate (6.8 %), excellent alkali stability, and good mechanical properties [28]. He's team used a novel tetraoctyloxycalix[4]arene (TQCX) to create continuous ion channels with lengths up to  $\sim 300 \text{ nm}$  in AEMs [29]. Compared with the pristine membrane, the hydroxide conductivity of the composite membrane increased by 70%, reaching the highest level of AEMs based on a polysulfone backbone....

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...It is reported that block copolymers [9], long side chain grafted polymers [33] and densely functionalized polymers [34] could benefit to the formation of the micro-phase separation. Besides, intermolecular interactions, e.g., hydrogen bonds and electrostatic interactions have been utilized to promote aggregation of cationic groups in order to obtain the continuous ion channels [35,36]. In addition, it is proved that introducing the bulky 3D structure into AEMs is a more facile way to improve their ionic conductivity [37]...

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